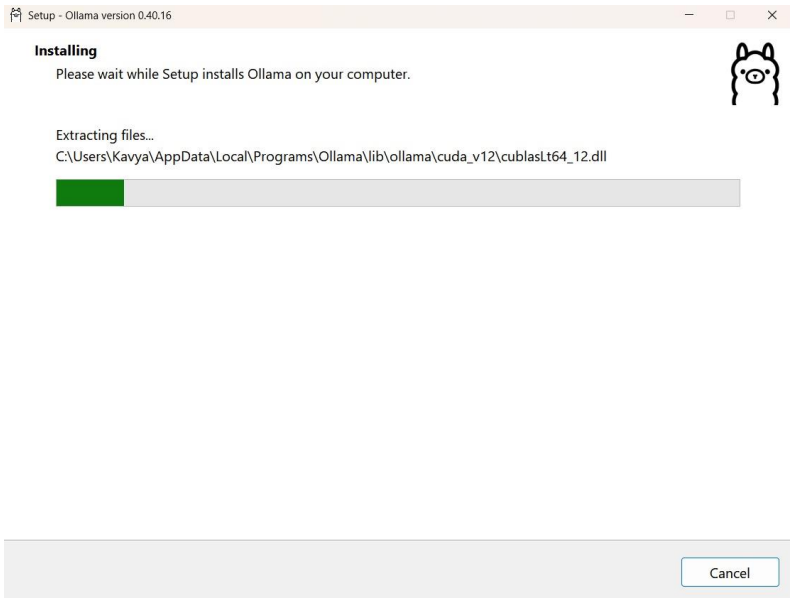
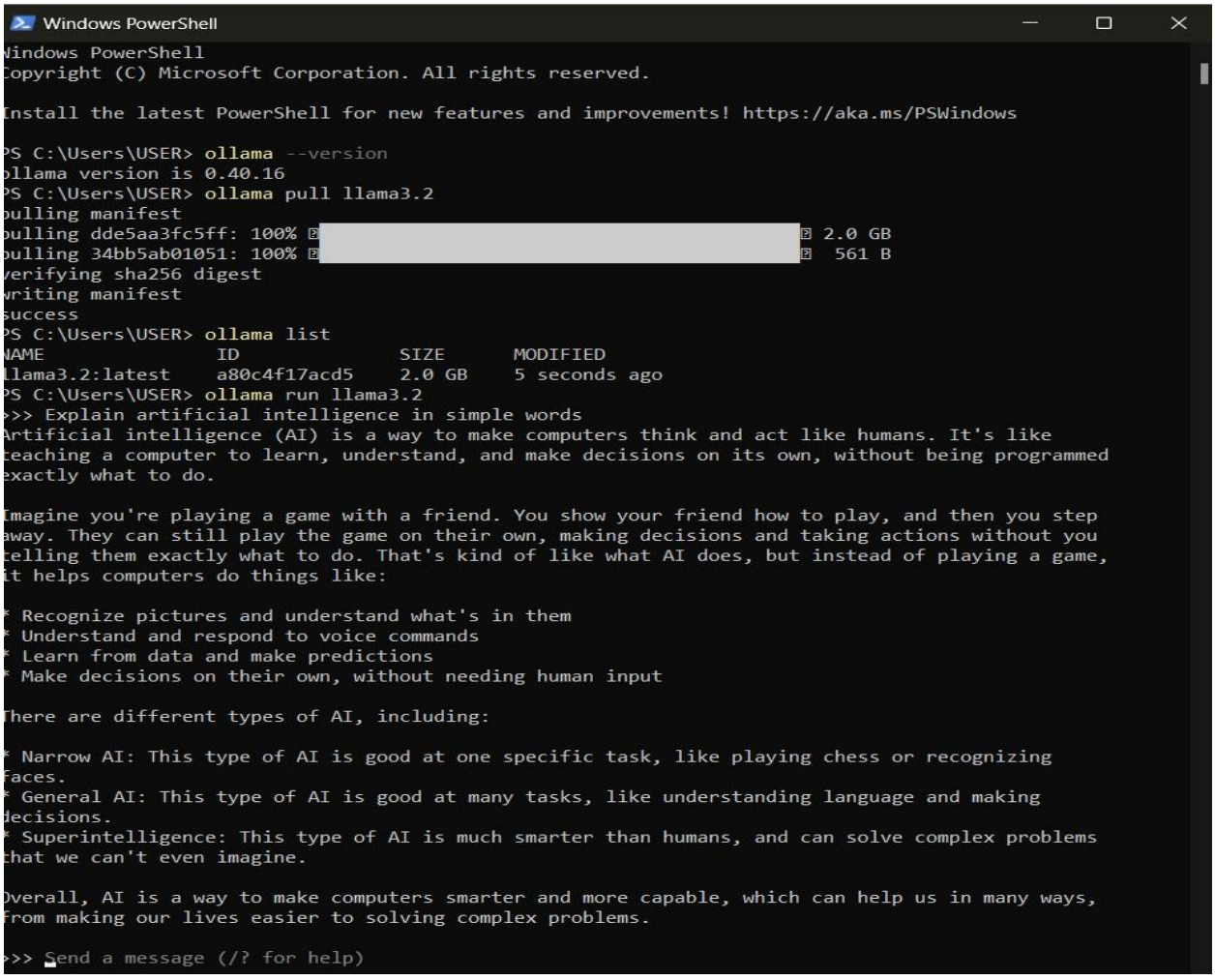
WEEK-2

2520090041

V.Pavani

Ollama Installation:




Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\USER> ollama --version
ollama version is 0.40.16
PS C:\Users\USER> ollama pull llama3.2
pulling manifest
pulling dde5aa3fc5ff: 100% 2.0 GB
pulling 34bb5ab01051: 100% 561 B
verifying sha256 digest
writing manifest
success
PS C:\Users\USER> ollama list
NAME ID SIZE MODIFIED
llama3.2:latest a80c4f17acd5 2.0 GB 5 seconds ago
PS C:\Users\USER> ollama run llama3.2
>>> Explain artificial intelligence in simple words
Artificial intelligence (AI) is a way to make computers think and act like humans. It's like
teaching a computer to learn, understand, and make decisions on its own, without being programmed
exactly what to do.
Imagine you're playing a game with a friend. You show your friend how to play, and then you step
away. They can still play the game on their own, making decisions and taking actions without you
telling them exactly what to do. That's kind of like what AI does, but instead of playing a game,
it helps computers do things like:
* Recognize pictures and understand what's in them
* Understand and respond to voice commands
* Learn from data and make predictions
* Make decisions on their own, without needing human input
There are different types of AI, including:
* Narrow AI: This type of AI is good at one specific task, like playing chess or recognizing
faces.
* General AI: This type of AI is good at many tasks, like understanding language and making
decisions.
* Superintelligence: This type of AI is much smarter than humans, and can solve complex problems
that we can't even imagine.
Overall, AI is a way to make computers smarter and more capable, which can help us in many ways,
from making our lives easier to solving complex problems.
>>> Send a message (/? for help)


Welcome
llm_client.py
main.py
test_client.py
requirements.txt
.env
llm_client.py > ...
from abc import ABC, abstractmethod
from google import genai
import requests
class LLMProvider(ABC):
@abstractmethod
def generate(self, prompt):
pass
class GeminiProvider(LLMProvider):
def __init__(self, api_key):
self.client = genai.Client(api_key=api_key)
def generate(self, prompt):
response = self.client.models.generate_content(
model="gemini-2.5-flash",
contents=prompt
)
return response.text
class OllamaProvider(LLMProvider):
def __init__(self, model="llama3.2"):
self.model = model
self.url = "http://localhost:11434/api/generate"
def generate(self, prompt):
response = requests.post(
self.url,
json={
"model": self.model,
"prompt": prompt,
"stream": False
},
timeout=120
)

Welcome | llm_client.py | main.py | test_client.py | requirements.txt | .env

main.py > ...

```python

import os
from dotenv import load_dotenv
from llm_client import GeminiProvider, OllamaProvider, LLMClient

load_dotenv()

provider_name = input(
    "Choose provider (gemini/ollama): "
).strip().lower()

if provider_name == "gemini":
    api_key = os.getenv("GEMINI_API_KEY")

    if not api_key:
        raise ValueError("Gemini API key not found in .env")

    provider = GeminiProvider(api_key)

elif provider_name == "ollama":
    provider = OllamaProvider("llama3.2")

else:
    raise ValueError("Invalid provider selected")

client = LLMClient(provider)

prompt = input("Enter your question: ")
answer = client.generate(prompt)

print("\nLLM Response:")
print(answer)

```

```
Enter your question: Explain deep learning in simple words

LLM Response:
Deep learning is a type of artificial intelligence (AI) that allows computers to learn from data without being explicitly programmed. Here's
 a simplified explanation:

**Imagine You're Trying to Recognize Cats**

1. You show a picture of a cat to a computer, and it's not sure what it is.
2. The computer looks at the picture and tries to find patterns, like:
       * The shape of the ears
       * The color of the fur
       * The shape of the nose
3. The computer uses these patterns to make a guess about what the picture is (in this case, a cat).
4. The computer keeps looking at more pictures of cats and tries to improve its guess.

This process is called "training" the computer. The more pictures of cats it sees, the better it becomes at recognizing them.
```

```
**The Result:**

Deep learning allows computers to learn complex patterns and make accurate predictions or decisions from data, like recognizing objects, spe
ech, images, or even predicting what might happen next in a movie. It's like a super-smart computer that can learn from experience and impro
ve over time!
PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\week-2>
PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\week-2> python -m unittest test_client.py
.
----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\week-2> python main.py
Choose provider (gemini/ollama):
```

```python
class GeminiProvider(LLMProvider):
    def __init__(self, api_key):
        self.client = genai.Client(api_key=api_key)

    def generate(self, prompt):
        response = self.client.models.generate_content(
            model="gemini-3.8-flash",
            contents=prompt
        )
        return response.text


class OllamaProvider(LLMProvider):
    def __init__(self, model="llama3.2"):
        self.model = model
        self.url = "http://localhost:11434/api/generate"
```

GeminiProvider Aa ab .* 1 of 1

Problems Output Debug Console Terminal Ports

```
> *"If it has four legs, fur, and whiskers, it is a cat."* (This fails often—what if the cat is missing a leg, or is a hairless cat?)

In **Deep Learning**, humans don't write the rules. We just feed the computer **millions of pictures** and tell it when it's right or wrong.
 Over time, the computer figures out its own rules.

### 3. Real-World Examples
You probably use deep learning every day:
*   **Face ID:** Your phone recognizing your face, even in the dark or if you put on glasses.
*   **Voice Assistants:** Siri and Alexa understanding your accents and slang.
*   **Self-Driving Cars:** Helping the car instantly tell the difference between a plastic bag blowing in the wind and a child walking into
the street.
*   **ChatGPT:** Learning the patterns of human language to write and chat like a person.

### In short:
**Deep learning is a way of making computers learn by example, using multi-layered math networks, so they can recognize patterns that are to
o complex for humans to write rules for.**
PS C:\Users\USER\OneDrive\Documents\Btech\Even sem -2\GEN-AI\week-2>
```